

Business statement & Executive summary

Business statement

The telecommunications company needs to reduce customer churn and protect recurring revenue by identifying the customer segments most likely to leave, understanding the service, contract, tenure, and billing factors associated with churn, and prioritizing customers for retention action. This project uses customer-level data, SQL analysis, a churn prediction model, and a Power BI dashboard to convert churn data into targeted retention decisions.

Business problems solved

1. Measured the scale of churn and revenue exposure

The analysis identified 7,043 customers, of which 1,869 churned. The overall churn rate is **26.54%**. Churned customers represent approximately **\$139,130.85 in monthly recurring revenue**.

2. Identified the contract types driving churn

Month-to-month customers have a churn rate of **42.71%**, compared with **11.27%** for one-year contracts and **2.83%** for two-year contracts. This highlights contract conversion as a major retention opportunity.

3. Identified early-tenure churn as an onboarding problem

Customers with 0–6 months of tenure have the highest churn rate at approximately **52.9%**, while customers with 49–72 months of tenure churn at approximately **9.5%**. This shows that the first months of the customer relationship are critical.

4. Identified high-risk payment methods

Customers using electronic checks have a churn rate of **45.29%**, substantially higher than customers using automatic bank transfer or credit card payments. This supports an autopay-adoption strategy.

5. Assessed the impact of service offerings on retention

Fiber Optic customers have a high churn rate of approximately **41.9%**. Customers without Online Security or Tech Support also have churn rates above **41%**, indicating an opportunity to use value-added service bundles to improve retention.

6. Built a churn prediction model for proactive action

A Logistic Regression model was developed with a **ROC-AUC score of 0.842**, showing good ability to distinguish likely churners from retained customers. The model identified tenure, contract type, Fiber Optic service, monthly charges, and total charges as important churn drivers.

7. Created a retention-priority customer list

The model classified **2,350 customers** as high risk, representing approximately **\$183,543.45 in predicted monthly revenue at risk**. These customers can be prioritized for retention outreach.

8. **Recommended customer-specific retention actions**

The project assigns actions such as annual-contract discounts, autopay enrollment, Tech Support bundles, Online Security bundles, and early-life onboarding calls.

Executive summary

Customer churn is a significant revenue-risk issue for the telecom business. Of 7,043 customers analyzed, 1,869 customers churned, resulting in an overall churn rate of 26.54%. The churned customer segment represents approximately \$139.1K in monthly recurring revenue.

The analysis shows that churn is concentrated among month-to-month customers, early-tenure customers, electronic-check users, Fiber Optic subscribers, and customers without Tech Support or Online Security. Month-to-month customers have a churn rate of 42.71%, compared with only 2.83% for customers on two-year contracts. Customers in their first six months have the greatest churn risk, showing that onboarding and early customer engagement require immediate attention.

A churn prediction model achieved a ROC-AUC score of 0.842 and identified 2,350 high-risk customers with approximately \$183.5K in monthly revenue at risk. The recommended retention strategy is to prioritize high-value, high-risk customers with annual-contract incentives, autopay offers, improved onboarding, and value-added support or security bundles.

The Power BI dashboard provides management with a single view of churn performance, customer segments, revenue exposure, and retention priorities, enabling data-driven actions to reduce churn and protect recurring revenue.